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State of Water Quality in the Creeks of Macquarie University

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Executive Summary

This report assesses the water quality in University Creek and Mars Creek, two freshwater streams that flow through the Macquarie University campus before entering the Lane Cove River, which is ecologically important and used for recreation. Monitoring was carried out to deliver a thorough evaluation of waterway condition, using ANZECC (2000) guidelines for freshwater systems and microbial contamination as benchmarks.

In situ field measurements (DO, pH, turbidity, temperature, TDS) and laboratory analyses (BOD, nitrate, phosphate, faecal coliforms) were conducted. Values for the Water Quality Index (WQI) were determined through the Weighted Arithmetic Mean Method (Brown et al., 1970).

Results showed that both creeks consistently ranked in the medium water quality category with WQI values between 50.2 and 64.8. With parameters like turbidity, TDS and pH staying within guideline thresholds, it suggests that baseline conditions are stable. Nonetheless, critical exceedances were noted for faecal coliforms, nutrient enrichment (nitrate, phosphate) and biological oxygen demand (BOD) with levels significantly exceeding acceptable limits. These stressors play a role in the reduction of dissolved oxygen levels and the moderation of overall creek health.

With a WQI of around 62, the Lane Cove River was also measured a medium range. This indicates that discharges from the campus contribute to catchment pressures in the broader area.

To protect ecological and recreational values, the report recommends that management should be taken to reduce microbial contamination, nutrient inputs and organic loading through infrastructure repairs, stormwater controls and riparian restoration.

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1. Introduction

Urban creeks play a vital role in linking constructed areas with natural watercourses. As noted by Simeonov et al. (2003), the decline in water quality of rivers and creeks is a worldwide problem associated with urban and industrial development, as well as natural phenomena like floods, droughts and bank erosion. These waterways are particularly vulnerable as they receive wastewater, stormwater and agricultural runoff but they are essential for drinking water, irrigation, industrial use, recreation and biodiversity.

The Water Quality Index (WQI), which combines various parameters into one score that can be compared to standards (Štambuk-Giljanović, 1999). Urban catchments are typically affected by stressors such as nutrient enrichment, faecal contamination and high organic loads, which are collectively referred to as the “urban stream syndrome” (Walsh et al., 2005).

This research applies the WQI method to University Creek and Mars Creek at Macquarie University, which both flow into the Lane Cove River.

The aim is to assess the ecological and recreational health of these waterways and identify key-stressors.

Objectives:

- Measure and analyse the key physical, chemical and biological water quality parameters at multiple sites.
- Suggest ways to enhance water quality management at Macquarie University and mitigate downstream effects on the Lane Cove River.

2. Methodology

2.1 Study Area:

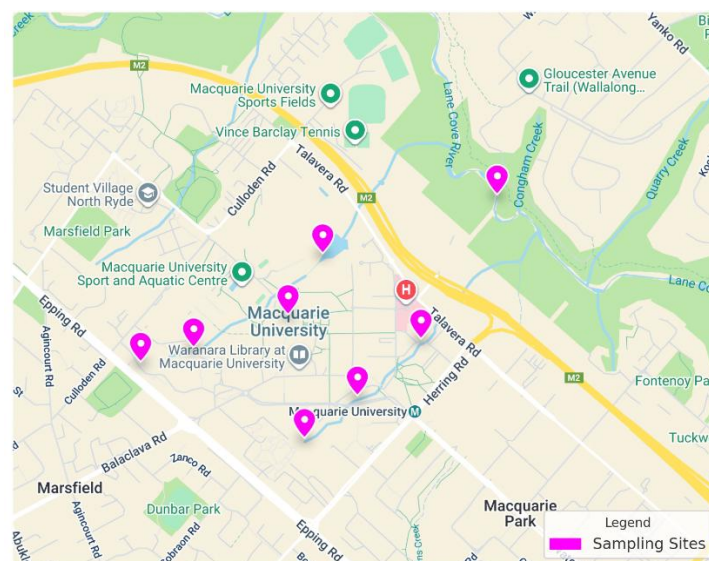


Figure 1: Sampling Sites Map (Source: Padlet)

Table 1: Sampling Area Location

Location	Latitude	Longitude
University Creek		
L1	-33.7784958	151.1130850
L2	-33.7768145	151.1155204
L3	-33.7749517	151.1184162
L4	-33.7691160	151.1211967
Mars Creek		
L1	-33.7756389	151.1062222
L2	-33.7750646	151.1084297
L3	-33.7756334	151.1062236
L4	-33.7698741	151.1217442

*Figure 2: Mars Creek Sampling Locations**Figure 3: University Creek Sampling Locations*

2.2 Sample Collection:

Water samples were collected in clean plastic containers. Before final collection, each container was rinsed with site water, then filled, closed tightly and properly labelled. Within a few hours, the parameters were determined in the laboratory.



Figure 4: Sample Collection

2.3 Field Measurements

Following parameters were measured in situ at the sampling locations using portable multiparameter probe (Hanna HI9829): pH, Dissolved oxygen (DO, % saturation), Temperature ($^{\circ}\text{C}$), Salinity ($\mu\text{S}/\text{cm}$), Turbidity (NTU) and TDS (ppm or mg/L).

These were recorded immediately in the field to avoid changes during storage.



Figure 5: Field Measurements

2.4 Laboratory Analyses:

Table 2: Laboratory Methods and Procedures

Parameter	Methods	Procedure
BOD	Real Tech BP Series BOD/COD meter (Real Tech, 2019)	Filling a quartz cuvette with the water sample. Then inserting it into the instrument after calibration and recording the displayed result once done.
Faecal coliforms	IDEXX Colilert Test	Add Colilert reagent to 100 mL sample, incubate at 35–37 °C for 24 h; yellow = total coliforms, fluorescence under UV = <i>E. coli</i> .
Nitrate/ Total N	Hach Cadmium Reduction Method 8171	Add NitraVer 5 reagent to 10 mL sample, shake, allow color to develop, measure spectrophotometrically.
Phosphate/ Total P	Hach PhosVer3 Ascorbic Acid Method 8048	Add PhosVer 3 reagent to 10 mL sample, shake, allow blue colour to develop, measure spectrophotometrically.



Figure 6: Laboratory Analysis

2.5 Data Analysis:

2.5.1 Water Quality Index (WQI): Weighted arithmetic means method (Brown et al., 1970).

Table 3: WQI Range and Quality

WQI Range	Quality
90-100	Excellent
70-90	Good
50-70	Medium
25-50	Bad
0-25	Very Bad

2.5.2 Benchmarking: Compared against ANZECC (2000) guidelines.

Parameter	Guideline
<i>Microbiological</i>	
Primary contact*	The median bacterial content in fresh and marine waters taken over the bathing season should not exceed 150 faecal coliform organisms/100 mL or 35 enterococci organisms/100 mL. Pathogenic free-living protozoans should be absent from bodies of fresh water.**
Secondary contact*	The median value in fresh and marine waters should not exceed 1000 faecal coliform organisms/100 mL or 230 enterococci organisms/100 mL.**
Nuisance organisms	Macrophytes, phytoplankton scums, filamentous algal mats, sewage fungus, leeches, etc., should not be present in excessive amounts.* Direct contact activities should be discouraged if algal levels of 15 000–20 000 cells/mL are present, depending on the algal species. Large numbers of midges and aquatic worms should also be avoided.

Figure 7: Summary of water quality guideline for recreational waters

Ecosystem type	Chl a	TP	FRP	TN	NO _x	NH ₄ ⁺	DO (% saturation) ^l		pH	
	(µg L ⁻¹)	(µg P L ⁻¹)	(µg P L ⁻¹)	(µg N L ⁻¹)	(µg N L ⁻¹)	(µg N L ⁻¹)	Lower limit	Upper limit	Lower limit	Upper limit
Upland river	na ^a	20 ^b	15 ^g	250 ^c	15 ^h	13 ⁱ	90	110	6.5	7.5 ^m
Lowland river ^d	5	50	20	500	40 ^o	20	85	110	6.5	8.0
Freshwater lakes & Reservoirs	5 ^e	10	5	350	10	10	90	110	6.5	8.0 ^m
Wetlands	no data	no data	no data	no data	no data	no data	no data	no data	no data	no data
Estuaries ^p	4 ^f	30	5 ^j	300	15	15	80	110	7.0	8.5
Marine ^p	1 ⁿ	25 ⁿ	10	120	5 ^k	15 ^k	90	110	8.0	8.4

Figure 8: Default trigger values for physical and chemical stressors for NSW

Ecosystem type	Salinity (μScm^{-1})	Explanatory notes
Upland rivers	30–350	Conductivity in upland streams will vary depending upon catchment geology. Low values are found in Vic. alpine regions ($30 \mu\text{Scm}^{-1}$) and eastern highlands ($55 \mu\text{Scm}^{-1}$), and high values ($350 \mu\text{Scm}^{-1}$) in NSW rivers. Tasmanian rivers are mid-range ($90 \mu\text{Scm}^{-1}$).
Lowland rivers	125–2200	Lowland rivers may have higher conductivity during low flow periods and if the system receives saline groundwater inputs. Low values are found in eastern highlands of Vic. ($125 \mu\text{Scm}^{-1}$) and higher values in western lowlands and northern plains of Vic ($2200 \mu\text{Scm}^{-1}$). NSW coastal rivers are typically in the range $200\text{--}300 \mu\text{Scm}^{-1}$.
Lakes & reservoirs	20–30	Conductivity in lakes and reservoirs is generally low, but will vary depending upon catchment geology. Values provided are typical of Tasmanian lakes and reservoirs.
Turbidity (NTU)		
Upland rivers	2–25	Most good condition upland streams have low turbidity. High values may be observed during high flow events.
Lowland rivers	6–50	Turbidity in lowland rivers can be extremely variable. Values at the low end of the range would be found in rivers flowing through well vegetated catchments and at low flows. Values at the high end of the range would be found in rivers draining slightly disturbed catchments and in many rivers at high flows.
Lakes & reservoirs	1–20	Most deep lakes and reservoirs have low turbidity. However, shallow lakes and reservoirs may have higher natural turbidity due to wind-induced resuspension of sediments. Lakes and reservoirs in catchments with highly dispersible soils will have high turbidity.
Estuarine & marine	0.5–10	Low turbidity values are normally found in offshore waters. Higher values may be found in estuaries or inshore coastal waters due to wind-induced resuspension or to the input of turbid water from the catchment. Turbidity is not a very useful indicator in estuarine and marine waters. A move towards the measurement of light attenuation in preference to turbidity is recommended.

Figure 9: Ranges of default trigger values for conductivity (EC, salinity) and turbidity

3. Result

3.1 University Creek:

Macquarie University Lake is a human made freshwater reservoir which built to manage floodwaters from Mars Creek. So, the ANZECC (2000) guideline values for freshwater lakes and reservoirs were utilized for water quality assessment.

Table 4: Measurements for University Creek

Factor	Location	Field Value	ANZEC C Guideline	Weight	Q Value	Weight * Q value	Compliance
DO (%)	L1	58.20%	90-110%	0.17	57	9.69	X
	L2	56.30%			56	9.52	X
	L3	56.30%			56	9.52	X
	Lane Cove River	58.20%			57	9.69	X
FC (fc/100 mL)	L1	53	<150 (primary contact)	0.16	52	8.32	✓
	L2	2000			18	2.88	X
	L3	>2000			17	2.72	X
	Lane Cove River	2000			18	2.88	X
pH	L1	7.48	6.5-8	0.11	92	10.12	✓
	L2	7.15			90	9.9	✓
	L3	8.4			67	7.37	✓
	Lane Cove River	7.6			92	10.12	✓
T (C) and change	L1	14.2 (2.75)		0.1	85	8.5	
	L2	13.4 (1.9)			90	9	
	L3	12.85 (1.35)			91	9.1	
	Lane Cove River	11.5 (0)			92	9.2	

Phosphate (mg/L)	L1	0.22		0.1	90	9	
	L2	0.51			70	7	
	L3	0.31			80	8	
	Lane Cove River	0.09			95	9.5	
Nitrates (m/L)	L1	4.4		0.1	80	8	
	L2	3.2			85	8.5	
	L3	2.2			87	8.7	
	Lane Cove River	0			100	10	
Turbidity (FNU/NT U)	L1	4.1	1-20	0.08	85	6.8	✓
	L2	3.4			90	7.2	✓
	L3	1.2			97	7.76	✓
	Lane Cove River	4.1			85	6.8	✓
TDS (ppm or mg/L)	L1	310		0.07	59	4.13	
	L2	302			60	4.2	
	L3	338			56	3.92	
	Lane Cove River	357			51	3.57	
BOD	L1	50.39		0.11	2	0.22	
	L2	50.59			2	0.22	
	L3	51.07			2	0.22	
	Lane Cove River	59.71			2	0.22	

Total P (mg/L))	L1	0.08	0.01				X
	L2	0.17					X
	L3	0.1					✓
	Lane Cove River	0.03					✓
Total N (mg/L))	L1	1	0.35				X
	L2	0.7					X
	L3	0.5					X
	Lane Cove River	0					✓
Salinity (uS/cm)	L1	620	20-30				X
	L2	604					X
	L3	676					X
	Lane Cove River	710					X

Table 5: University Creek WQI

Location	WQI	Quality
L1	64.78	Medium
L2	58.42	Medium
L3	57.31	Medium
lane cove river	61.98	Medium

3.2 Mars Creek:

Mars Creek is an urban freshwater creek. The ANZECC (2000) guideline values for lowland rivers were used to evaluate its water quality.

Table 6: Measurements for Mars Creek

Factor	Location	Field Value	ANZEC C Guideline	Weight	Q Value	Weight * Q value	Compliance
DO (%)	L1	48.00%	85-110%	0.17	45	7.65	X
	L2	27.60%			20	3.4	X
	L3	72.40%			73	12.41	X
	Lane Cove River	59.90%			58	9.86	X
FC (fc/100 mL)	L1	>2000	<150 (primary contact)	0.16	17	2.72	X
	L2	>2000			17	2.72	X
	L3	>2000			17	2.72	X
	Lane Cove River	>2000			17	2.72	X
pH	L1	7.17	6.5-8	0.11	90	9.9	✓
	L2	6.86			87	9.57	✓
	L3	7.34			91	10.01	✓
	Lane Cove River	7.3			91	10.01	✓
T (C) and change	L1	15.72 (3.48)		0.1	74	7.4	
	L2	17.05 (4.81)			72	7.2	
	L3	15.35 (3.11)			80	8	
	Lane Cove River	12.24 (0)			92	9.2	

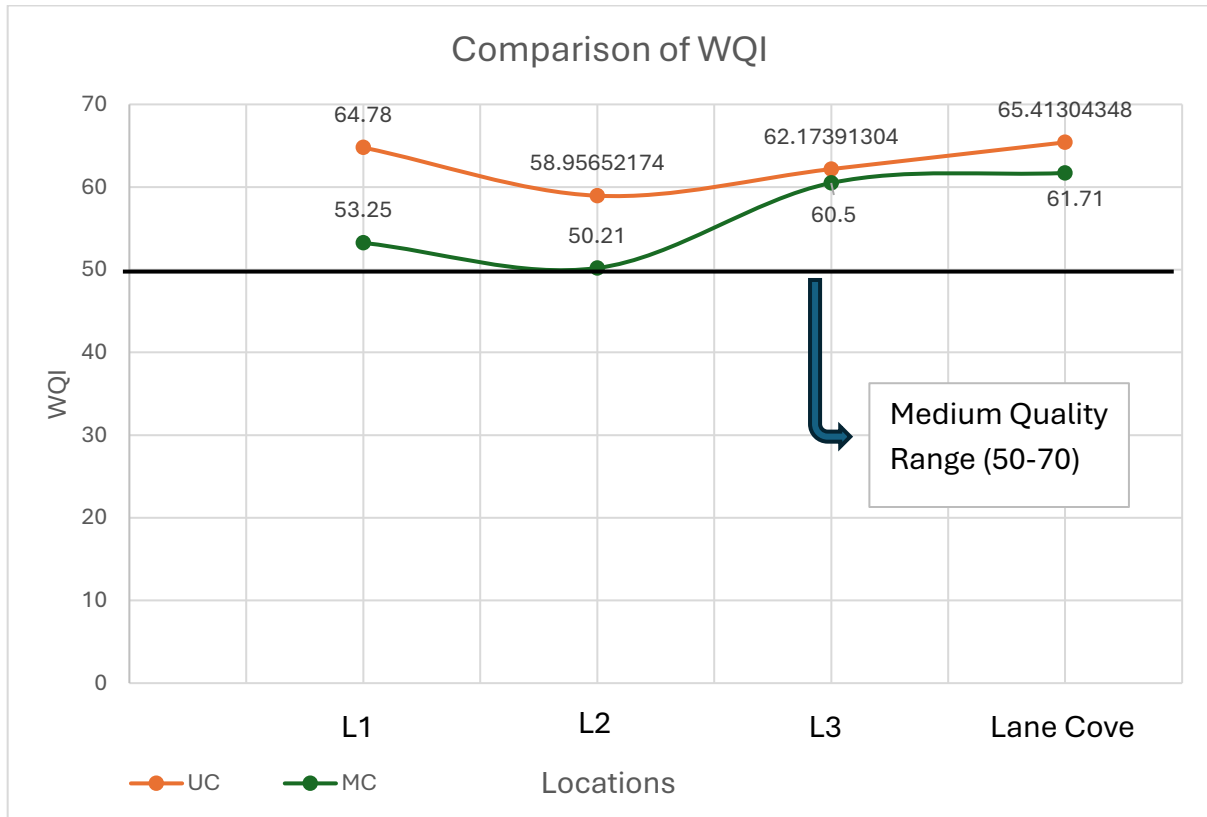
Phosphate (mg/L)	L1	0.03		0.1	100	10	
	L2	0.31			90	9	
	L3	0.49			70	7	
	Lane Cove River	0.09			92	9.2	
Nitrates (m/L)	L1	0.5		0.1	98	9.8	
	L2	0.4			99	9.9	
	L3	0.4			99	9.9	
	Lane Cove River	0.8			95	9.5	
Turbidity (FNU/NT U)	L1	31.6	6-50	0.08	52	4.16	✓
	L2	5.8			85	6.8	X
	L3	5.5			86	6.88	X
	Lane Cove River	3.4			92	7.36	X
TDS (ppm or mg/L)	L1	778		0.07	20	1.4	
	L2	627			20	1.4	
	L3	389			48	3.36	
	Lane Cove River	366			52	3.64	
BOD	L1	68.11		0.11	2	0.22	
	L2	59.01			2	0.22	
	L3	59.1			2	0.22	
	Lane Cove River	66.36			2	0.22	

Total P (mg/L))	L1	0.08	0.05				X
	L2	0.17					X
	L3	0.1					X
	Lane Cove River	0.03					✓
Total N (mg/L))	L1	1	0.5				X
	L2	0.7					X
	L3	0.5					✓
	Lane Cove River	0					✓
Salinity (uS/cm)	L1	620	125- 2200				✓
	L2	604					✓
	L3	676					✓
	Lane Cove River	710					✓

Table 7: Mars Creek WQI

Location	WQI	Quality
L1	53.25	Medium
L2	50.21	Medium
L3	60.5	Medium
lane cove river	61.71	Medium

4. Discussion



The analysis of the Water Quality Index (WQI) indicates that University Creek and Mars Creek are categorized in the medium quality range (50-70), with WQI scores between 50.2 and 64.8. Even though no sites were found as poor or very poor, but the findings suggest that the creeks are experiencing pressure from multiple primary stressors.

4.1 Essential Factors Influencing Medium Quality:

- Faecal coliform levels were invariably higher than the NHMRC primary contact guideline of 150 cfu/100 mL and at multiple locations which surpassing 2000 cfu/100 mL. This suggests microbial contamination because of leaking sewage systems, stormwater inflows or animal waste. Increased microbial counts present health dangers for recreational use.
- The levels of nutrients (nitrates and phosphates) surpassed the ANZECC suggested values which is indicating contributions from fertilisers, organic matter or urban runoff. These nutrients can increase the risk of eutrophication that led to algal blooms and subsequent oxygen depletion.
- The values for Biochemical Oxygen Demand (BOD) were very high (50–70 mg/L), greatly surpassing the guideline limit of <4 mg/L. This shows a high level of organic loading, which depletes dissolved oxygen as microorganisms consume oxygen to decompose organic matter (Allan, 2004; Chapman, 1996).

- Dissolved oxygen (DO) levels were low (56–58% saturation in University Creek and as low as 27% at Mars Creek (L2), which is likely a consequence of high BOD and nutrient enrichment.

4.2 Parameters That Fall Within the Guideline Ranges:

Conversely, temperature, turbidity and total dissolved solids (TDS) typically complied with ANZECC standards which indicates that sediment loading, and physical water clarity are not the primary concerns there. These creeks are more susceptible to erosion.

4.3 Comparison Between Creeks:

Although both creeks received a “medium” range, University Creek generally exhibited slightly higher Water Quality Index values (57–65) than Mars Creek (50–60). This indicates that Mars Creek is affected to a greater extent, especially at L1 and L2 where DO and faecal coliforms is the biggest issues. This might indicate varying hydrological conditions, catchment pressures or flow regimes between the two creeks.

4.4 Downstream Effects:

Lane Cove River, which receives discharges from both creeks, showed a medium quality as well (WQI ~62). This indicates that the water issues in the river are being exacerbated by pollution from university creeks. As the Lane Cove River flows through Lane Cove National Park, persistent problems with bacteria and nutrients could negatively impact the river's flora and fauna and make it unsafe for recreation.

5. Recommendations

Microbial Source Tracking: Examine the faecal contributions from humans and animals. Fix leaking sewage systems and stormwater sewage cross connections.

Management of Nutrients: Decrease the use of fertilizer on campus grounds. Introduce biofiltration swales and riparian buffers.

Organic Load Reduction: Implement gross pollutant traps and debris management systems to decrease the amount of organic matter entering creeks.

Habitat Restoration: Bolster the riparian vegetation to boost natural filtration and bank stabilization.

Monitoring Program: Maintain long term WQI monitoring and add parameter-specific testing in accordance with ANZECC guidelines.

Community Engagement: Increase awareness among staff, students and residents to encourage sustainable practices and safeguard campus waterways.

6. Conclusions

The water quality of University Creek and Mars Creek is currently moderate, as indicated by WQI values ranging from 50 to 65. Although pH, turbidity, TDS, and temperature are acceptable, the health of these waterways is affected by microbial pollution (faecal coliforms), enrichment with nutrients (phosphate and nitrate), elevated organic loading (BOD) resulting in low levels of dissolved oxygen. These stressors threaten aquatic ecosystems and recreational safety, with downstream effects on the Lane Cove River.

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